

BVSP – Sprint Build Specification (SBS)

Sprint 0 – Application Foundation

Version: 1.0

Sprint Duration: Initial Build

Document Status: Approved

Project: Bhanap Vaiyaktika Seva Portal (BVSP)

Sprint Objective

Build the complete application foundation, project architecture, design system, Firebase integration, routing, authentication framework, and reusable components.

No business functionality should be implemented in this sprint.

The output of this sprint should be a production-ready application shell that is ready for feature development.

Sprint Deliverables

Google AI Studio must generate:

- Complete React project
- TypeScript configuration
- Firebase integration
- Firestore service layer
- Authentication framework
- Responsive layout
- Navigation
- Reusable UI components
- Empty pages
- Theme
- PWA configuration
- Error handling
- Loading framework

Project Structure

Generate the following structure:

```
src/
├── assets/
├── components/
├── config/
├── contexts/
├── firebase/
├── hooks/
├── layouts/
├── pages/
├── services/
├── styles/
├── types/
├── utils/
├── App.tsx
└── main.tsx
```

Every folder must contain appropriate starter files.

Pages to Generate

Public

- Home
- About
- Programmes
- Programme Details
- Resources
- FAQ
- Important Dates
- Login

Authenticated

- Dashboard
- My Seva
- Daily Logger
- Progress
- Summary

Coordinator

- Coordinator Dashboard

Super Admin

- Admin Dashboard

All pages should use the common layout and navigation.

No business logic yet.

Routing

Implement React Router with protected route support.

Public routes must be accessible without login.

Authenticated routes should redirect unauthenticated users to Login.

Role-based route guards should support:

- Sadhaka
 - Coordinator
 - Super Admin
-

Layout

Generate:

- Header
 - Navigation Bar
 - Footer
 - Page Container
 - Responsive Mobile Menu
 - Breadcrumb Component
 - Page Title Component
-

Navigation

Desktop

- Home
- Programmes
- Resources
- FAQ
- About
- Login

Mobile

Hamburger menu

Responsive drawer

Theme

Generate a reusable design system.

Colours

Primary

Deep Teal

Secondary

Temple Gold

Accent

Saffron

Background

Warm Ivory

Typography

Large readable fonts

Rounded cards

Soft shadows

Minimal animation

Components

Generate reusable components:

- Button
 - Card
 - PageHeader
 - SectionTitle
 - SearchBox
 - LoadingSpinner
 - ErrorCard
 - EmptyState
 - ConfirmationDialog
 - Modal
 - Badge
 - DataTable
 - Pagination
 - Toast Notifications
-

Firebase

Create

firebase.ts

Initialize

- Authentication
- Firestore
- Storage

Use environment variables.

No secrets inside code.

Firestore Service Layer

Generate empty services:

AuthService

ProgrammeService

ResourceService

UserService

SankalpaService

DailyLogService

ReportService

SettingsService

Each service should include placeholder CRUD methods.

Pages must call services—not Firestore directly.

Authentication Framework

Implement:

Authentication Context

Protected Routes

Role-based Authorization

Session Persistence

Logout

Do not implement login UI logic yet.

PWA

Configure

- Manifest
- Icons
- Install prompt
- Offline shell
- Service worker

Error Handling

Generate:

404 Page

Error Boundary

Friendly error messages

Fallback UI

Loading

Generate reusable:

Loading Spinner

Skeleton Loader

Progress Indicator

Accessibility

Support:

Keyboard navigation

High contrast

Large touch targets

Semantic HTML

Screen readers

Responsive Design

Support:

Mobile

Tablet

Desktop

No page should require horizontal scrolling.

Coding Standards

TypeScript Strict Mode

Reusable functions

Reusable components

Service layer pattern

No duplicate code

No hardcoded values

Configuration-driven approach

Deliverables

At the end of Sprint 0, the project should:

- Build successfully
 - Run without errors
 - Connect to Firebase
 - Support routing
 - Support authentication framework
 - Support role-based navigation
 - Display all empty pages
 - Be ready for Sprint 1 development
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Explicitly Out of Scope

Do **not** implement:

- User Registration
- Login verification
- Sankalpa submission
- Daily Logger functionality
- Reports
- Dashboard analytics
- Firestore CRUD operations
- Resource uploads
- Admin workflows

Only create the framework and placeholders.

Acceptance Criteria

Sprint 0 is complete when:

- Application builds without errors.
- All routes are operational.
- Responsive navigation works.
- Firebase is initialized correctly.

- Authentication framework is in place.
 - Reusable components are available.
 - Empty pages render consistently.
 - Theme is applied across the application.
 - Code follows the TIG and BRB.
 - The project is ready for Sprint 1 without structural changes.
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Developer Instructions (to include with Google AI Studio)

Use this short prompt with the three documents:

You are a senior React, TypeScript, Firebase, and UI/UX engineer. Build BVSP Sprint 0 exactly according to the attached Business Requirements Blueprint (BRB), Technical Implementation Guide (TIG), and Sprint Build Specification (SBS). Generate production-quality, modular, maintainable code. Follow the Firestore schema in the TIG. Do not invent business functionality beyond Sprint 0. Where implementation is intentionally deferred, add clear TODO comments rather than placeholders or assumptions.

One strategic suggestion

Because your target is to launch the **Goregaon Sabha pilot**, I would actually label this as **Sprint 0 – Foundation & Public Experience** and include the **Home page, About, Programmes, Resources, FAQ, and Important Dates with real UI** rather than leaving them empty. That gives you a working portal almost immediately, while the authenticated features (registration, Sankalpa, Daily Logger, reports, and admin) can follow in Sprint 1. It will make reviews easier and let stakeholders see tangible progress much earlier.